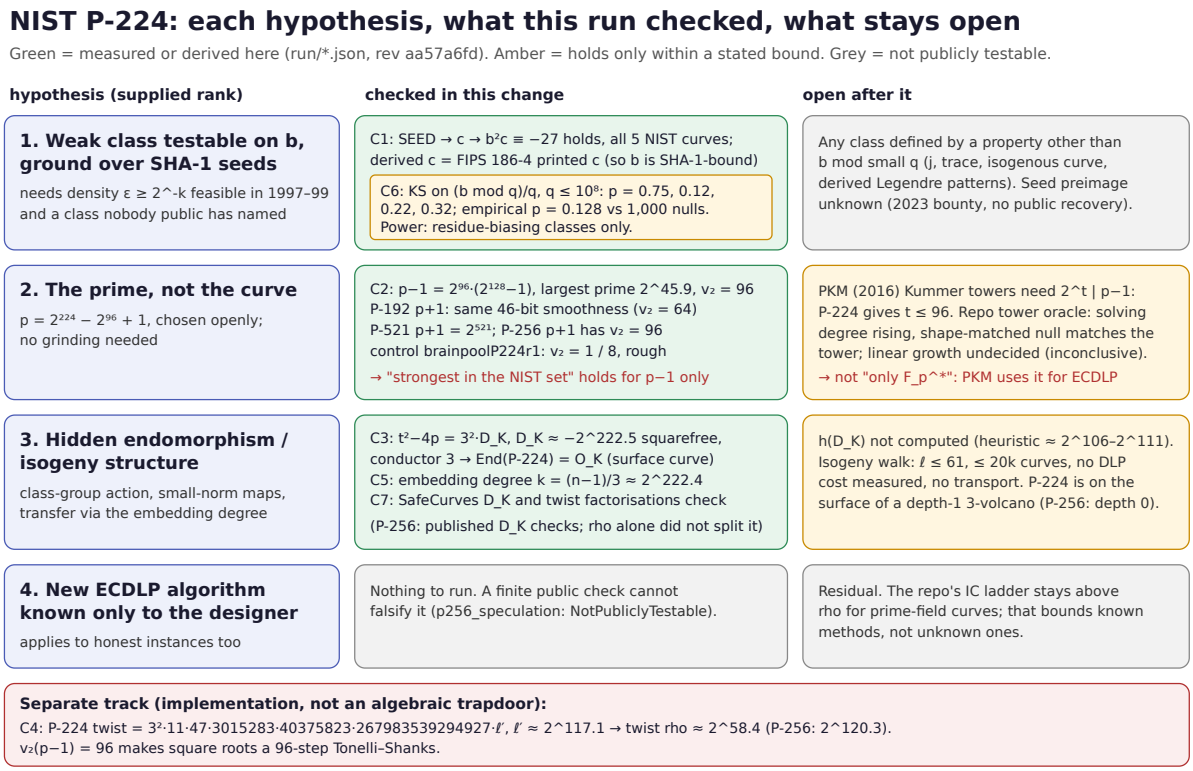
NIST P-224: a ranked hidden-weakness hypothesis list, checked against this repository

2026-10-08

Status

Assessment plus one native run; no ECDLP cost measured. Protocol: PROTOCOL.md, frozen in commit `aa57a6fd` before the run. Run: run/, produced by `target/release/p224_hypotheses` at that commit, with a clean tree, rustc 1.97.0 and 4 threads, in 122 s of wall time. Every check finished, no stop condition fired, and the b-residue test raised no flag.

This is not an index-calculus measurement. It has no rho ratio, no phase is priced, and no scoreboard row is owed (`AGENTS.md` §7a). Under `AGENTS.md` "Intermediate-cost and asymmetric-access hypotheses", a hypothesis about a designer's capability is not evidence that a trapdoor exists. A bounded null is not nonexistence.



Each hypothesis, what this run checked, what stays open.

Evidence labels used below:

- **MEASURED**: computed by this run (file named).
- **DERIVED**: algebra from stated inputs.
- **VERIFIED-PUBLISHED**: a published factorisation re-checked here by product and primality, not found here.
- **PRIOR**: an earlier committed result in this repository.
- **OPEN**: not established.

The supplied claim

The supplied text ranks four hypotheses "by how compatible each is with the public evidence and 1999-era compute":

1. a weak class recognisable from `(p, a, b)` without point counting, found by grinding SHA-1 seeds. It cites `b_seed_profile` and the Solinas-correlation experiment as nulls;
2. the prime, not the curve. It says hidden-SNFS primes (Fried–Gaudry–Heninger– Thomé) have no ECDLP analogue. It calls P-224's 46-bit-smooth `p − 1` "the strongest structure of this kind in the NIST set", affecting "only DLP in F_p^*, not the curve group";
3. hidden endomorphism or isogeny structure. It says all public checks pass (embedding degree, anomaly, CM discriminant and class number "enormous") and that the isogeny walk found no weak curve at small degree;
4. an unpublished ECDLP algorithm: the unfalsifiable residual.

The ranking itself is an interpretation. This report checks the factual claims it rests on, for P-224.

Claim-by-claim

#	claim in the supplied text	status for P-224	evidence
1a	b is bound to a SHA-1 seed, so a grind is the only freedom	**holds.** SEED → c (X9.62 A.3.3.1) gives FIPS 186-4's printed c and $b^2c \equiv -27$ for all five NIST curves	MEASURED run/seeds.json
1b	the b-residue KS profile came back null	**was P-256 only, at $q \le 10^5$**, not the 10^6 its module doc describes. P-224 is now null at every bound (table below)	MEASURED run/b_residues.json
1c	statistical probes on b are the right kind of test	**partly.** The KS test sees only classes that bias b mod q. A cheaply testable class defined any other way is invisible to it	DERIVED, PROTOCOL §4
1d	the Solinas-correlation experiment came back null	**P-256 only; not a test on b; miscalibrated.** It tests bit correlations of the P-256 reduction map. Its z uses $\sqrt{(E(1-E)/n)}$ with E = p_r p_c, not the independence variance $\sqrt{(p_r(1-p_r)p_c(1-p_c)/n)}$. That deflates every z by $\approx \sqrt{3}$: the recorded max 2.46 over 131,072 pairs sits far below the ≈ 4.5 a calibrated null gives. Rescaled (≈ 4.3) it is still below its 6.1 threshold, so the null probably survives with less power than claimed. Not rerun here	code read src/cryptanalysis/solinas_correlations.rs, PRIOR RESEARCH_P256.md
2a	hidden-SNFS trapdoor is real for F_p DLP	**holds**, with a date fix: ePrint 2016/961, EUROCRYPT 2017. A Solinas prime is *openly* SNFS-friendly, so no hiding is involved; for P-224 the F_p^* DLP is also trivial by Pohlig–Hellman (largest prime 2^45.9)	literature; MEASURED run/prime_sides.json
2b	P-224's p − 1 is 46-bit smooth	**holds.** $p - 1 = 2^{96} \cdot 3 \cdot 5 \cdot 17 \cdot 257 \cdot 641 \cdot 65537 \cdot 274177 \cdot 6700417 \cdot 67280421310721$, all proved prime, $v_2 = 96$	MEASURED run/prime_sides.json
2c	"the strongest structure of this kind in the NIST set"	**true for p − 1 only.** P-192's $p + 1 = 2^{64}(2^{128} - 1)$ has the same 46-bit smoothness; P-521's $p + 1 = 2^{521}$ is 2-smooth; P-256's p + 1 has $v_2 = 96$ and largest prime 2^93.2	MEASURED run/prime_sides.json
2d	it "only affects DLP in F_p^*, not the curve group"	**too strong.** Petit–Kosters–Messeng (PKC 2016), the one published algebraic ECDLP mechanism for prime fields, builds factor bases from towers whose Kummer form needs 2^t | p − 1. P-224 supports $t \le 96$; every other NIST prime gives 1. Its Dickson-torus form needs 2^t | p + 1: P-192 (64), P-256 (96), P-521 (521). Nikolaev (CTCrypt 2024) flags P-224 under the PKM criterion. This repository's tower-oracle rounds measured a rising solving degree, with a shape-matched null matching the tower and linear growth undecided. That is an open lead, not an attack	PRIOR research/notes/index-calculus/RESEARCH_PKM_TOWER_ORACLE.md, research/notes/cm-isogeny/RESEARCH_PKM_CRITERION.md
3a	huge embedding degree	**holds, exactly.** k = (n − 1)/3 ≈ 2^222.4, from a complete factorisation of n − 1. P-256: k = (n − 1)/3	MEASURED run/embedding_degree.json
3b	not anomalous	**holds** ($n \ne p$)	DERIVED
3c	CM discriminant enormous	**holds, and is now resolved here.** $t^2 - 4p = -3^3 \cdot 29 \cdot 79 \cdot 7523 \cdot 40927 \cdot 11549194661 \cdot L_{148}$, factored completely. So the conductor of Z[π] is 3 and $D_K = -3 \cdot 29 \cdot 79 \cdot 7523 \cdot 40927 \cdot 11549194661 \cdot L_{148} \approx -2^{222.5}$, equal to SafeCurves' value. The isogeny walk places P-224 on the 3-volcano's surface, so End(P-224) = O_K. That closes the walk README's End(E) = unk for P-224. For P-256, rho alone did not split the cofactor; SafeCurves' factorisation checks, giving conductor 1 and End = O_K	MEASURED run/cm_discriminant.json; VERIFIED-PUBLISHED run/published_factorizations.json
3d	class number enormous	**heuristic, not computed.** $h(D_K) = \sqrt{\|D_K\|} \cdot L(1,\chi)/\pi \approx 2^{109.6} \cdot L(1,\chi)$; a GRH-conditional asymptotic lower bound gives about 2^106	OPEN
3e	isogeny walk found no weak curve at small degree	**holds only within its bounds**: $\ell \le 61$, 20,000 curves, a fixed trait list, no DLP cost measured, no discrete-log transport. It omits that P-224, unlike P-256, sits on a depth-1 3-volcano	PRIOR research/isogeny_walk_20261004/README.md

#	claim in the supplied text	status for P-224	evidence
4	an unpublished algorithm is not publicly testable	**holds**; nothing was run	PRIOR `src/cryptanalysis/p256_speculation.rs`

The b-residue profile (C6)

KS statistic of $(b \bmod q)/q$ against Uniform[0,1) over primes $5 \le q \le Q$. The empirical p-value comes from 1,000 SHA-1-derived null coefficients mod the P-224 prime at $Q = 10^6$ (`run/b_null_panel.json`).

curve	Q	primes	D	asymptotic p	empirical p
P-224	10^5	9,590	0.006894	0.752	—
P-224	10^6	78,496	0.004220	0.122	0.128
P-224	10^7	664,577	0.001295	0.215	—
P-224	10^8	5,761,453	0.000400	0.315	—
P-256	10^5	9,590	0.011490	0.159	—
P-256	10^6	78,496	0.002673	0.629	0.627
P-256	10^7	664,577	0.001058	0.446	—
P-256	10^8	5,761,453	0.000297	0.691	—

- **Reproduction.** The P-256 $Q = 10^5$ row reproduces the recorded $D = 0.011490$, $p = 0.1589$.
- **Calibration gate (PROTOCOL §4): passes.** 48 of 1,000 null p-values fall below 0.05, inside the 30–70 band. 12 fall below 0.01, 2 below 0.001, and the mean is 0.498.
- **Flag: none.** No asymptotic $p < 0.001$, and P-224's statistic sits inside the null panel.
- **Cost.** $Q = 10^8$ took seconds, not the "hours" the `b_seed_profile` module doc predicts (`run/run.json`: 4.4 s for all eight rows).

Separate track: implementation, not an algebraic trapdoor

These do not bear on hypotheses 1–4 and do not refute them (`AGENTS.md`).

- **Twist (C4).** $\#E' = 3^2 \cdot 11 \cdot 47 \cdot 3015283 \cdot 40375823 \cdot 267983539294927 \cdot \ell'$ with $\ell' \approx 2\text{^}117.1$ a probable prime. Expected rho in the largest twist subgroup is about 2^58.4, matching SafeCurves. P-256's twist gives about 2^120.3. An x-only implementation that skips point validation therefore leaks P-224 keys at about 2^58 work. This repository's `research/p224_attack/` demonstrates the related invalid-curve route. This closes the walk README's "212-bit composite, unfactored" twist cofactor.
- **Square roots.** $v_2(p - 1) = 96$ forces a 96-step Tonelli–Shanks (or equivalent) for point decompression. P-224 is the only NIST prime with $v_2(p - 1) > 1$.

Interpretation (labelled, not a finding)

The run does not change the supplied ranking's order. It changes two of its statements:

- hypothesis 2's smooth-prime structure is not unique to P-224's $p - 1$;
- that structure is the stated prerequisite of a published ECDLP approach, PKM, whose status in this repository is inconclusive rather than closed.

Hypothesis 2 for P-224 is therefore an open, testable lead, not a dead end. Its next step is the tower-oracle programme at 2-power Kummer depth matched to P-224, with the full-cost and matched-rho gates of `AGENTS.md` §§1–8.

Hypothesis 1's grind bound is a scenario, not evidence. Koblitz–Menezes (ePrint 2015/1018) argue that a seed-ground weak class large enough to find would be too large to have escaped public notice. Bernstein et al. (BADA55, ePrint 2014/571) show how much freedom a seed procedure leaves a designer. The seed preimages remain unpublished: the 2023 Valsorda bounty reports no recovery in the sources checked.

Requirement status

item	status
check each factual claim of the supplied ranking for P-224	verified complete (table above)
run P-224 counterparts of the cited P-256 probes	b-residue: complete. Solinas correlation: **not attempted for P-224**; its calibration defect is reported and the module is unchanged
close End(E) and twist gaps of the isogeny-walk README	complete for P-224; P-256 End resolved only via a verified published factorisation

item	status
class number of D_K	open
PKM cost at P-224 tower depth	open (not attempted; needs its own protocol)
hypothesis 4	not testable

Reproduce

```
cargo test --release --lib cryptanalysis::p224_hypotheses
cargo build --release --bin p224_hypotheses
target/release/p224_hypotheses --out /tmp/p224-check   # compare with run/
```

SHA-256 of the committed outputs:

```
0dfebc6b32941387b7978b8c7c11fe1b8f575630c22557e5fe9a928f5b51e712  seeds.json
f9a6d3e006c699a38fcfe79018932b59cc9cf1bc16168eeea844596753444f6b  prime_sides.json
c6d8efa5c9960122501dd38248c149b344d98ae21b1b9f80b13ef751e9147204  cm_discriminant.json
4fa39800f2da9704af99aa2a24099c4344509c74dfe1473d1f72365793b09870  twist.json
796b2251d31082e6f3aafeb136c6bfe6f063593d79921b45693db35b7088210c  embedding_degree.json
bd6d2284fd5a05e4063c5ebf5bb17ed6202afb5ef9cb40b267e46b339542ee4b  published_factorizations.json
d09798a42e36d7cca27cf5647c230920fce9d4032539af3aa18fda20ecd1117d  b_residues.json
6e4dafc81ec747328a67d3ef82d34972e5835193f2b6dea18a8e8e44405b00f5  b_null_panel.json
```

`run.json` holds wall times, so it differs between runs. The other files are deterministic.

Sources

- NIST, FIPS 186-4, Appendix D: seeds, `c`, curve parameters.
- Bernstein and Lange, SafeCurves: `disc.html` and `twist.html`, fetched 2026-10-08.
- Fried, Gaudry, Heninger and Thomé, A kilobit hidden SNFS discrete logarithm computation, EUROCRYPT 2017.
- Koblitz and Menezes, A Riddle Wrapped in an Enigma.
- Bernstein et al., How to manipulate curve standards (BADA55).
- Petit, Kosters and Messeng, *Algebraic approaches for the elliptic curve discrete logarithm problem over prime fields*, PKC 2016.
- Nikolaev, *On the correctness of criterion for resistance to Petit–Kosters–Messeng method*, CTCrypt 2024.
- Valsorda, NIST curve seeds bounty, 2023.